

The house price-rent ratio: evidence from new measures

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Journal of
Financial
Economic Policy

749

Abstract

Purpose – This study aims to determine whether there is a long-run relationship between house prices and rents in US cities. This long-run relationship has been implied by formal asset-pricing models but has often been difficult to find empirically.

Design/methodology/approach – The author examines this question with a brand-new data set on house prices and rents compiled by researchers at the Philadelphia Fed, which has overcome some previous measurement problems of previous indices, among them an underestimate of rent growth. The author also allows for structural breaks in the relationship, while previous studies had imposed parameter constancy.

Findings – While previous, flawed measures indicated a large difference in geographical dispersion between house prices and rents, the new indices show only a slight difference in such spatial variation. Importantly, while using standard testing methods yields little evidence of a long-run relationship between prices and rents for most cities, allowing for breaks with the Lee–Strazicich test reveals such a long-run relationship in most markets.

Practical implications – Given the existence of a long-run relationship, high levels of home prices relative to rents are still indicative of housing market froth.

Originality/value – To the best of the author's knowledge, this is the first study on this topic that uses this new data set, which has been specifically developed to overcome the deficiencies of previous house price and rent indices. It is also among very few that allow for structural breaks in the price-rent relationship, which is important given the numerous changes in the US housing market over recent decades, such as securitization and more widely available credit.

Keywords Housing supply and demand, House prices, rents, Financial economics, Asset pricing, Time-series models

Paper type Research paper

Received 7 November 2024
Revised 24 July 2025
Accepted 25 July 2025

Introduction

The value of one's home represents the largest financial asset for most households, and fluctuations in housing values lead to gains and losses for banks and other lenders and investors in mortgage-backed assets. Moreover, if house prices exceed levels consistent with fundamentals, not only does it impair affordability but also can lead to a financial bubble. Once the bubble bursts, a financial crisis can ensue and impact the economy with harsh recessions (see [Jorda et al. \(2015\)](#) on how credit-fueled housing bubbles lead to financial crises and economic contractions).

One way to gauge excess in a given housing market is with the ratio of house prices to rents. Theoretical models imply prices and rents should have a stationary long-run relationship. A bubble may be indicated by either deviations from the long-run relationship or an inability to find such a relationship at all. A number of studies have been conducted on the topic ([Andre et al., 2014](#); [Gallin, 2008](#); [Hansen et al., 2024](#); [Mekhid and Zemcik, 2009](#); [Tang et al., 2020](#)); all of which examine this issue for the USA.



JEL classification – G12, R30, R31

Conflict of interest: The author reports no conflict of interest.

Journal of Financial Economic
Policy
Vol. 18 No. 5, 2026
pp. 749-771
© Emerald Publishing Limited
1757-6385
DOI 10.1108/JFEP-11-2024-0343

Despite the predictions of theory, a stationary relationship has often proved hard to find empirically. A possible obstacle to a stationary long-run relationship may be the fact that measured house prices have seemingly become more geographically dispersed compared to rents ([Tang et al., 2020](#)). [Yao \(2021\)](#) also finds prices are more varied across the USA than rents. She develops a model in which the greater land required for a house leads to this higher dispersion between rents and prices. This difference in dispersion means that the price-rent ratio is not a good indicator of housing bubbles.

One issue in examining the relationship between house prices and rents (their relative dispersion and whether they share a long-run relationship) is the availability of reliable data. [Agorastos et al. \(2024\)](#) develop a data set of house prices and rents for 30 US cities based on newly digitized newspaper ads going back to 1890. The authors note that quality-adjusted rent data did not exist for the USA before 2000. In addition to a number of other issues the authors address with their data, they show that rents have, contrary to some previous research, risen more than thought. Thus, prices and rents across the USA may not differ in dispersion by as much as [Tang et al. \(2020\)](#) and [Yao \(2021\)](#) found using data going back only to 1975 and 1980, respectively. And if differences in geographic dispersion are not as stark as previously found, there could be a stationary relationship between the two.

We find that while house prices are currently more varied across the USA than rents, this difference is very much less than that found in [Tang et al. \(2020\)](#) and [Yao \(2021\)](#). Indeed, in years past, rents in some periods were more dispersed than house prices. Given this feature of the data, we then seek to determine whether there is a long-run stationary relationship between prices and rents. Using standard unit root tests, including the Zivot–Andrews method, which allows for a break under the alternative of stationarity, we generally do not find such a relationship. However, given the changes in USA housing over more than a century, we use the Lee–Strazicich test, which allows for structural change under both the null and alternative hypotheses. With this test, we find the majority of cities display a long-run relationship between home values and rents.

We do note that while some models cited above imply a bivariate stationary relationship between house prices and rents, there are others ([Himmelberg et al., 2005](#)) which imply a relationship in which the price-rent ratio is the inverse of the user cost of owning a house. The user cost entails interest rates, a risk premium, tax considerations and the expected capital gain from owning a home. One implication of this model is that there should be an inverse relationship between the real long-term interest rate and the price-rent ratio (see also [Amaral et al., 2024](#)). While we cannot perform a formal test of the user cost theory with this data, we will investigate the relationship between interest rates and the price-rent ratio.

This paper proceeds as follows. The second section describes the previous literature. The third section explains the data and methodology. The fourth section describes our results, and the fifth section concludes.

Previous literature

The notion of a long-run relationship between home prices and rents arises from the idea that asset prices should equal the present value of the cash flows the asset generates. [Campbell and Shiller \(1987\)](#) developed a model with this implication. Empirically, when applied to equities, the model has been interpreted as implying that stock prices and dividends ought to have a stationary relationship or be cointegrated. When applied to the housing sector, the model has been taken to mean that house prices and rents should have a stationary relationship.

[Gallin \(2008\)](#) cites an asset-pricing theory to motivate a study on whether prices and rents have a stationary relationship in the USA. The author obtains data on national house prices

from the Freddie Mac index and rents from the consumer price index (CPI). His analysis does indicate that prices and rents in the USA are cointegrated. [Mekhid and Zemcik \(2009\)](#) examine not national but metro-level data on prices and rents for 23 US cities. Like Gallin, these authors use rent data from the CPI (“Rent of Primary Residence”) but get their house price data from the office of federal housing enterprise oversight index. The authors use the cross-sectionally augmented IPS panel unit root test and a panel cointegration test but cannot reject the null of no stationary relationship between prices and rents with either method.

[Andre et al. \(2014\)](#) investigate the price-rent ratio for the USA and 15 other organization for economic cooperation and development countries. These authors examine fractional integration – they allow their estimate of the dynamics of the ratio to indicate stationary, nonstationary or explosive behavior. For the USA, the price-rent ratio actually appears explosive. [Tang et al. \(2020\)](#) examine the price-rent relationship for 300 cities in the USA. The authors, like [Mekhid and Zemcik \(2009\)](#), apply the CIPS panel unit root test to the ratio and cannot reject the null of a unit root.

A somewhat different approach is taken by [Hansen et al. \(2024\)](#). These authors examine the price-rent ratio in 382 US cities with the Freddie Mac house price index and rent data from the Bureau of Labor Statistics (BLS). These authors test with the [Phillips et al. \(2015\)](#) method, which posits a null hypothesis of nonstationarity and an alternative hypothesis of explosive behavior. This method does not allow for the possibility of a stationary relationship between prices and rents. The authors find, in the majority of cases, cities exhibit explosive behavior over the COVID-19 years.

Thus, especially at the city level, a stationary relationship has proved elusive, despite the predictions of asset-pricing theory. There are other models, however, that suggest prices and rents are affected differently by fundamentals, and thus the price-rent ratio may not be a reasonable indicator of housing market conditions. [Yao \(2021\)](#) notes that home prices are more geographically dispersed than rents. In particular, the author obtains house price data for 181 US cities from Freddie Mac and rent data from the Census and Community Survey, and shows that in 1980, the coefficient of variation (CV) across the USA was 0.18 for rents and 0.33 for home prices. The difference in dispersion grew – by 2010 the CV for rents had increased to 0.27, but that for home prices had risen to 0.62. Tang, Zheng and Zhu also show that prices are more dispersed than rents and that this disparity has increased through time. The authors indicate that the CV of home values was 1.94 times that of rents in 1984, but this multiple grew to 2.75 by 2007. [Van Nieuwerburgh and Weill \(2010\)](#) and [Gyourko et al. \(2013\)](#) also present theories implying home values should be more dispersed than rents.

[Yao \(2021\)](#) develops a model to explain the greater geographical dispersion of house prices relative to rents. In the authors’ model, houses prices take up more land per housing unit than rental apartments. So long as there are differences in land values across cities, home values will exhibit more variation than rents because of the greater use of land in building houses. She calibrates her theoretical model and finds it can explain much of the regional dispersion in home values and rents. She also notes that the model implies that the increase in the price-rent ratio can be explained by fundamentals, which casts doubt on the use of the ratio as a bubble indicator.

Yao’s model can indeed have relevance for the dynamics and dispersion of prices and rents. [Davis and Heathcote \(2007\)](#) note that a house is a bundle of two goods – the structure and the land. These authors stated that between 1975 and 2006, real land values nearly quadrupled, while structure values only went up 33% – so, in light of Yao’s model, as apartments take up less land, they should rise less than houses. With similar reasoning, [Bostic et al. \(2007\)](#) develop the idea of land leverage. Like Davis and Heathcote, Bostic, Longhofer and Redfearn note that a house is a bundle of two goods – the structure and the

land. The authors state that construction costs for the structure should be uniform within a particular urban housing market. The authors' land leverage hypothesis states that if construction costs are indeed uniform across housing markets, then differences in house price appreciation across regions must arise from asymmetric exposure to common shocks to land values. Again, given Yao's focus on how apartments take up less land than houses, Bostic, Longhofer and Redfearn's conjecture is consistent with the idea that house prices could possibly become more geographically dispersed than rents.

In assessing the empirical work on the price-rent relationship, there are several issues that may have affected results to this point. First, to the extent that the price-rent ratio reflects housing market conditions, it does so in the long run (no theory implies it must be constant at all times), and it may require a long span of data to detect such a relationship, if it exists. [Ambrose et al. \(2013\)](#), for instance, do find a long-run relationship between prices and rents in the city of Amsterdam but use 355 years of data to do so.

The data used in studies for the USA has been over much shorter periods. [Gallin \(2008\)](#) uses data between 1970 and 2005, while [Mekhid and Zemcik \(2009\)](#) use price and rent indices spanning 1975–2006. [Andre et al. \(2014\)](#) have data from 1970 to 2011. [Tang et al. \(2020\)](#) have a data set covering 1975–2017. [Hansen et al. \(2024\)](#) use data starting in 1977 and ending in 2021. [Yao \(2021\)](#) uses data spanning 1980–2010.

Obtaining data going back prior to the 1970s has been challenging. [Agorastos et al. \(2024\)](#) note that there is a national index for US house prices going back to 1890 developed by Robert Shiller, who “spliced together several data sources to obtain a national housing sales price series” ([Agorastos et al., 2024](#), p. 2). The authors point out that there are no city-level annual price indices available for the US prior to 1975. In addition, there are no quality-adjusted rent indices at the city level before 1975.

[Agorastos et al. \(2024\)](#) develop their house price and rent indices by obtaining data on digitized newspaper ads for 30 US cities – yielding 27 million observations over 1890–2006. Their house price data is fairly similar to those developed by Case and Shiller and the Federal Housing Finance Administration subsequent to 1990, but it differs in earlier years. The authors note that Shiller's price series relies on a single survey for the 1890–1934 years and truncated loan data for 1953–1975. As a result, [Agorastos et al. \(2024\)](#) “find evidence of a sizeable interwar housing cycle that is not present in the Shiller series” (p. 4). Measured rents, as calculated by previous methods for the BLS and the CPI, have been subject to criticism for downward bias. [Gordon and Van Goethem \(2007\)](#) and [Crone et al. \(2010\)](#) both present results indicating that rents measured for the CPI are biased downward for reasons such as nonresponse bias. [Agorastos et al. \(2024\)](#) cite this research and find calculating rents based on newspaper ads leads to higher estimates than those from the CPI measures (note that this research implies that the CPI has risen more, and living standards have risen less over the years than official statistics suggest).

[Yao \(2021\)](#) obtains rent data from the US Census and American Community Survey. [Tang et al. \(2020\)](#) obtain rent data from the Fair Market Rent database compiled by the Department of Housing and Urban Development. All other papers on the price-rent relationship for the USA ([Gallin, 2008](#); [Mekhid and Zemcik, 2009](#); [Andre et al., 2014](#); [Hansen et al., 2024](#)) use data from the BLS or directly from the CPI, which is of course subject to the criticism of downward bias.

The data developed by [Agorastos et al. \(2024\)](#) indicates, consistent with the results of [Gordon and Van Goethem \(2007\)](#) and [Crone et al. \(2010\)](#), that rents have risen faster than previously believed. And as rents have risen more than previously thought, they may have also become more dispersed than previously believed; that is, the difference in the geographical dispersion between rents and prices may not be as great as found in [Tang et al. \(2020\)](#) and

Yao (2021). If the difference in dispersion is not as great between prices and rents, then there could be a long-run relationship between the two. We will thus investigate the issue with the new data set.

Data and methodology

Given the aforementioned deficiencies in historic price and rent indices, Agorastos *et al.* (2024) examine digitized newspaper listings for 2.7 million home and rent prices from 30 cities spanning 1890–2006 and, with hedonic methods, construct annual house value and rent indices. Their data is made available at: www.philadelphiafed.org/surveys-and-data/regional-economic-analysis/historical-housing-prices.

As noted, the authors find differences with this data compared to that from previous sources. House prices in the USA rose faster than previously believed over 1965–1995, for instance. In addition, previous research had suggested that rents had fallen in the USA over the 20th century, whereas this new data set indicates they rose. Rents thus may or may not be as lacking in geographical dispersion as Tang *et al.*'s (2020) and Yao's (2021) analyses would indicate.

We thus calculate, as did the aforementioned authors, the coefficients of variation (CV) over time for rents and prices across the 30 cities. The data set created by Agorastos *et al.* (2024) did not have prices and rents for all 30 cities for the entire 1890–2006 period (Table 1 displays the data available for all cities). There are 13 cities for which there is rent and price data available for all years over 1890–2006, while in the other 17, prices and rents are only available for certain years. We thus calculate two sets of CVs for rents and prices – one “balanced” sample for the 13 cities with complete rent and price data over the entire span and 1 “unbalanced” for all cities. These will be calculated for each year over 1890–2006.

Next, to obtain a purely stylized gauge of comovement between rents and prices, we will compute, for each city, 10-year rolling correlations between rent and prices. We will observe how this measure of comovement has changed through time, as well as the average of the rolling correlation values across different cities.

We next will formally investigate whether there is a stationary, long-run relationship between prices and rents. Ideally, we would use cointegration methods. While some previous studies have tested for stationarity by testing for unit roots on the price-rent ratio, this imposes a particular cointegrating relationship between prices and rents (depending on how the price and rent indices are calculated, testing for a unit root in the ratio is tantamount to imposing a cointegrating vector of $[1, -1]$). It will thus be desirable to apply either univariate or panel cointegration tests to rents and prices, but this would necessitate both rents and prices having unit roots. We must therefore test for unit roots in each city's rent and price series. To anticipate our results, we will find that for most cities, house prices do appear to have unit roots, while rents in most markets appear stationary. Thus it will not be possible to use cointegration methods. We therefore must, as has been done in previous studies, apply unit root tests to the price-rent ratio for each city.

Given that unit root tests are often plagued by low power, we will first apply panel unit root tests (as did Mekhid and Zemcik (2009) and Tang *et al.* (2020)) to the price-rent ratios of the 30 cities. Panel unit root tests typically posit a null hypothesis that all series in the panel have unit roots, but they differ in the alternative hypotheses. The Levin–Lin–Chu test, for instance, specifies as the alternative hypothesis that all series are stationary. This is, of course, a highly restrictive alternative hypothesis. The Im, Pesaran and Shin (IPS) test has the same null hypothesis as the Levin–Lin–Chu technique, but the alternative is that at least one series is stationary. In addition, the Levin–Lin–Chu test requires a balanced panel, while the IPS test does not. The IPS test has also been modified to correct for cross-sectional

Table 1. Sample data

City	Home prices	Rents	City	Home prices	Rents
Atlanta	1890–2006	1890–2006	Miami	1910–2005	1915–2006
Baltimore	1908–2006	1890–2006	Minneapolis	1890–2006	1890–2006
Boston	1890–2006	1890–2006	Nashville	1890–2006	1890–2006
Charleston	1911–2006	1894–2006	New Orleans	1893–2006	1890–2006
Chicago	1890–2006	1890–2006	New York	1890–2006	1890–2006
Cincinnati	1890–2006	1890–2006	Philadelphia	1890–2006	1891–2006
Cleveland	1890–2006	1894–2006	Phoenix	1910–2006	1910–2006
Dallas	1890–2006	1890–2006	Pittsburgh	1890–2006	1892–2006
DC	1890–2005	1890–2005	Portland	1898–2006	1898–2006
Detroit	1890–2006	1890–2006	Saint Louis	1890–2006	1890–2006
Houston	1900–2005	1896–2005	Salt Lake City	1890–2006	1891–2006
Las Vegas	1943–2006	1948–2005	San Diego	1890–2006	1907–2006
Los Angeles	1890–2006	1890–2006	San Francisco	1890–2006	1890–2006
Louisville	1890–2006	1890–2006	Seattle	1910–2006	1910–2006
Memphis	1890–2006	1891–2006	Tampa Bay	1905–2006	1915–2006

Note(s): These are the dates of coverage from the Philadelphia Federal Reserve Bank’s Historical House Prices Project at www.philadelphiafed.org/surveys-and-data/regional-economic-analysis/historical-housing-prices#data-download

Source(s): Author’s own work

dependence among the series, a problem which, if unaddressed, leads to size problems (over-rejection of the null) in testing. We will thus apply the Levin–Lin–Chu and cross-sectionally augmented Im, Pesaran and Shin (CIPS) tests to the 13 cities where price and rent data are available over the full 1890–2006 sample (the balanced panel). We will, in addition, apply the CIPS test to all 30 cities where price and rent data availability varies across the 1890–2006 time span.

As the results of panel unit root tests do not inform as to how many or which cities may have a stationary price-rent relationship, we will also perform univariate unit root tests on the ratio for each city. Given the increase in the price-rent ratio over time (see Supplementary Figures 1–30), we include trends in the ratios. Although the presence of trends requires the inclusion of a trend term in the test equations, these terms can lower the power of these tests. We thus also apply the Dickey–Fuller Generalized Least Squares (DF-GLS) test, which detrends the series before the test is applied, potentially increasing power. However, to anticipate our results, we will fail to reject the null of a unit root in the price-rent ratio for most cities with the augmented Dickey–Fuller (ADF), Phillips–Perron (PP) and DF-GLS tests. This result does not necessarily imply that prices and rents lack a long-run relationship. The testing procedures just mentioned all impose constant parameters for the entire sample. The housing market, of course, has been through numerous changes over the more than century-long sample. The way home purchases are financed is one example (Miles, 2019). If the price-rent relationship has undergone changes, failure to allow for structural breaks in the ratio could lead to failure to reject the null of a unit root even if there is a long-run relationship between the two. We will thus test for unit roots in the ratio with methods that allow for breaks. The Zivot–Andrews test has been widely applied, so we will use it. However, this technique only allows for a break under the alternative hypothesis of stationarity. No break is allowed under the null of a unit root. This test thus tends to wrongly identify the date of the break (Lee and Strazicich, 2013). Moreover, the Zivot–Andrews test statistic is calculated based on the estimated break date, leading to problems of inference.

We find that results from the Zivot–Andrews test generally indicate the lack of a long-run relationship between house prices and rents, as is the case with the ADF, PP and DF-GLS tests. Given the problems with the Zivot–Andrews test, we go on to use the Lee–Strazicich method, which allows for breaks under both the null and alternative hypotheses, for the 30 city price-rent ratios. All tests are conducted using the EViews statistical software package.

Results

Figure 1 displays the annual rolling coefficients of variation for the entire (unbalanced) 30-city sample. Note that most price and rent indices were normalized to 1 in 1948; hence, the relatively low values for the CV for both rents and prices in that year. As displayed in Table 2, at the start of the sample in 1890, the coefficient of variation for rents in this unbalanced sample was actually higher than that for prices. Home prices did start to exhibit greater geographical dispersion than rents starting in 1978. By 2006, the CV of home values in this sample is 0.575108, while that for rents is 0.433145. Thus, house price dispersion for all 30 cities is 1.327 times that of rent dispersion by the end of the sample. So, like Tang *et al.* (2020) and Yao (2021), home values exhibit greater variation at the end of the sample than do

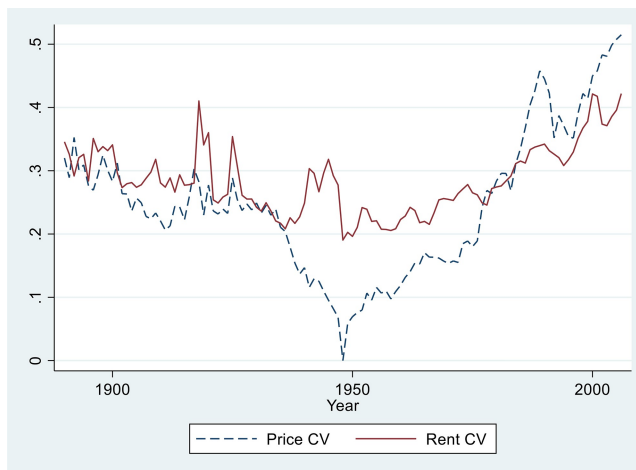


Figure 1. Unbalanced (full sample) rolling coefficient of variation

Note(s): These are the rolling coefficients of variation for all 30 cities in the sample

Source: Author's own work

Table 2. Coefficients of variation for home prices and rents

	Balanced sample		Full sample	
Housing variable	1890 coefficient of variation	2006 coefficient of variation	1890 coefficient of variation	2006 coefficient of variation
Home prices	0.29605	0.575108	0.320038	0.51489
Rents	0.305512	0.433145	0.345649	0.421895

Source(s): Author's own work

rents. However, the difference found by both authors in geographical variation between prices and rents is much greater than what we have found. Yao calculates that prices had 1.833 times the geographic variation as rents in 1980, and that by 2010 this figure had increased to 2.296. Tang *et al.* (2020) with their sample, found prices had 1.84 times the dispersion of rents in 1984, and by 2017 the number was 2.75. We find, as did Yao and Tang *et al.*, that price dispersion has risen faster than rent dispersion – again, in our sample, prices began exhibiting greater cross-city variation than rents starting in 1978 – but again, the difference is not nearly as stark as that found by previous authors.

We also show, in Figure 2, the coefficient of variation for the balanced sample of 13 cities in which prices and rents are available for the entire 1890–2006 sample. Rents again exhibit slightly more dispersion than prices in 1890. For years, the level of dispersion fluctuates, and rent variation is sometimes greater than price variation, and vice versa. Home price dispersion does start to grow more than rent dispersion, and, by 1972, the former is higher than the latter. By 2006, the house price CV is 0.575108, while that for rents is 0.433145. Thus, price variation is 1.22 times rent variation, again a difference much smaller than that found by Yao or Tang *et al.*

We next examine 10-year rolling correlations between rents and prices in each of the 30 cities. For brevity of presentation, we only display graphs for the three most populous metro areas – New York, Los Angeles and Chicago – in Figures 3–5. These three are representative of those of the other cities; however, there is no trend in correlation – it goes up and down and is on average positive but sometimes negative and again does not display a pronounced sample-wide positive or negative trend. One qualification to this attribute of time-varying correlation is that toward the end of the sample, in 19 of 30 cities, there was a pronounced drop in comovement toward 2006. This may reflect the housing bubble in the lead-up to the Great Financial Crisis, when house prices rose rapidly to bubble levels.

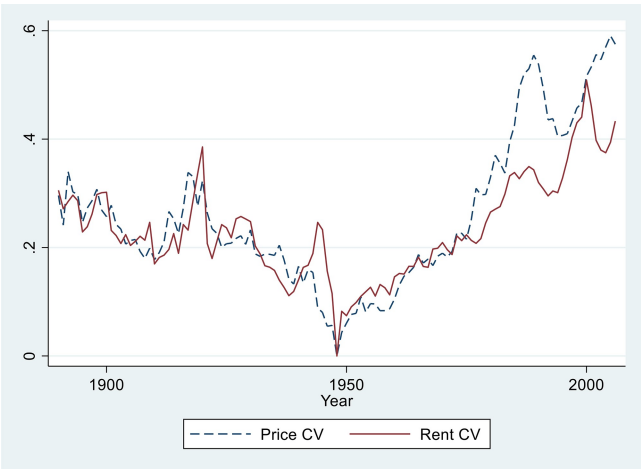


Figure 2. Balanced (13 cities) rolling coefficient of variation

Note(s): These are the coefficients of variation for the 13 cities (Atlanta, Boston, Chicago, Cincinnati, Dallas, Detroit, Los Angeles, Louisville, Minneapolis, Nashville, New York, San Francisco and Saint Louis) for which data is available over the full 1890–2006 sample

Source: Author’s own work

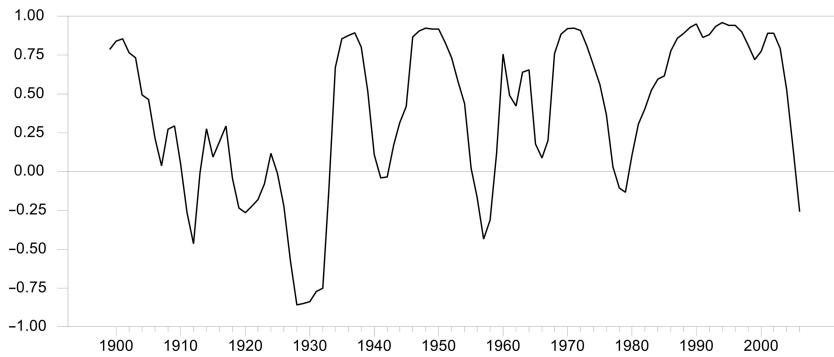


Figure 3. New York City price/rent rolling correlation
Source: Author's own work

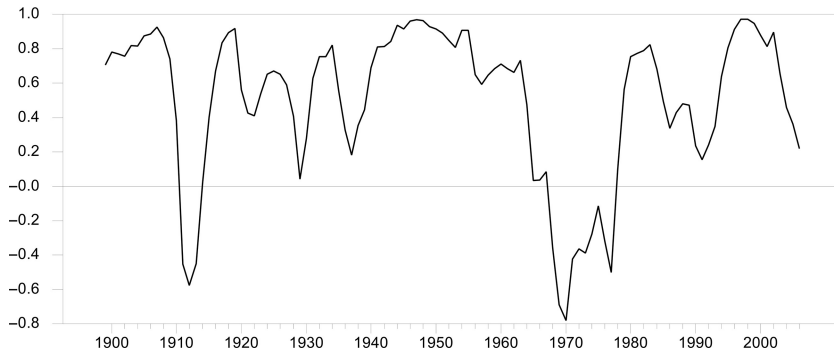


Figure 4. Los Angeles price/rent rolling correlation
Source: Author's own work

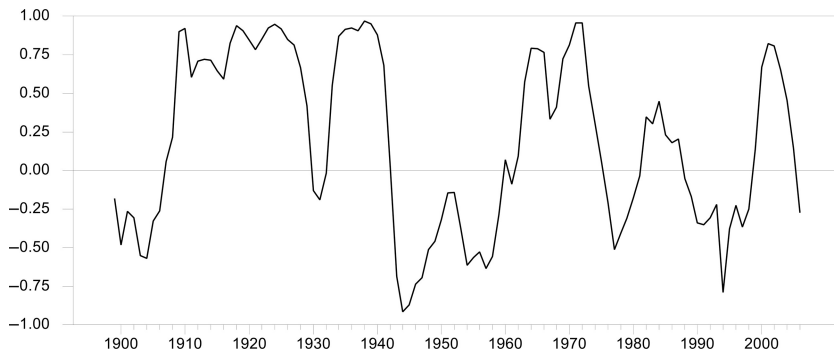


Figure 5. Chicago price/rent rolling correlation
Source: Author's own work

Table 3. Rolling correlation averages

City	Average	City	Average
Atlanta	0.468193	Miami	0.669667
Baltimore	0.220831	Minneapolis	0.276503
Boston	0.586801	Nashville	0.422686
Charleston	0.297482	New Orleans	0.323877
Chicago	0.187286	New York	0.375843
Cincinnati	0.408027	Philadelphia	0.46798
Cleveland	0.303423	Phoenix	0.592708
Dallas	0.584131	Pittsburgh	0.407594
DC	0.219483	Portland	0.372622
Detroit	0.465721	Saint Louis	0.291897
Houston	0.408272	Salt Lake City	0.33785
Las Vegas	0.453924	San Diego	0.392741
Los Angeles	0.506522	San Francisco	0.422059
Louisville	0.318651	Seattle	0.481082
Memphis	0.516468	Tampa Bay	0.422997

Note(s): These numbers are the averages of the 10-year rolling correlations between house prices and rents calculated for each city
Source(s): Author’s own work

There is also much variation in correlation across cities. Table 3 displays the averages for the rolling correlations over 1890–2006 for the 30 cities. Just as there was great variation through time, there is also variation in comovement across cities. Miami, Phoenix and Boston have the highest average correlations, with values of 0.669, 0.592 and 0.586, respectively. The cities with the lowest average correlations between prices and rents are Baltimore, DC and Chicago (values of 0.22, 0.219 and 0.187, respectively). We note that all average values of correlation are positive. There is no discernible geographic pattern to the correlations. There are several southern cities (Atlanta, Memphis, Miami, Nashville and Tampa) with higher average correlations than the majority of metro areas, while there are others (Charleston and New Orleans) with lower correlations than most. Similarly, there are East Coast cities such as Boston and Philadelphia which display high correlations, but others, such as New York and Baltimore, exhibit generally low comovement between rents and prices. On the west coast, Los Angeles and San Francisco have relatively high levels of rent-price correlation, while San Diego and Portland exhibit relatively low rent-price comovement.

We also examined whether there might be a connection between the elasticity of housing supply and rent-price correlations. We relied on the housing elasticity supply measure of Saiz (2010), but we found the correlation between this measure and rent-price correlation was insignificant (the correlation was -0.00635). While this exercise seems to show that rents and prices appear to comove positively, if imperfectly, within housing markets, the exercise is purely descriptive. Any interpretation would rightfully be very tentative, as most prices and some rents are, as we will discuss, nonstationary, making measured correlations questionable. We thus begin formal analysis to determine the extent to which there is a long-run relationship between rents and prices.

To determine if there is such a long-run relationship, we must first determine the stationarity properties of rents and prices. If both rents and prices are nonstationary, there still may be a stationary relationship between them. As mentioned, other studies have tested for whether the price-rent ratio has a unit root, but this imposes a specific cointegrating vector on the relationship. Ideally, we would like to test for cointegration between the two variables,

which does not require such a restriction, but this requires both rents and prices to be integrated to the same degree. We thus test for both unit roots.

Table 4 displays ADF and PP test results for unit roots for home prices. As displayed, we fail to reject the null of a unit root in most cases. With the ADF, we can reject at the 5% level in six cases and at the 10% level in two others. With the PP method, we reject at 5% for two cities and at 10% for four. Overall, house prices appear to be nonstationary in most cases.

Table 4. ADF and PP tests on the house prices

City	ADF test result	PP test result	City	ADF test result	PP test result
Atlanta	Fail to reject	Fail to reject	Miami	Fail to reject	Fail to reject
Baltimore	Fail to reject	Fail to reject	Minneapolis	Fail to reject	Fail to reject
Boston	Fail to reject	Fail to reject	Nashville	Fail to reject	Fail to reject
Charleston	Fail to reject	Fail to reject	New Orleans	Fail to reject	Fail to reject
Chicago	Fail to reject	Fail to reject	New York	Fail to reject	Fail to reject
Cincinnati	Reject	Fail to reject	Philadelphia	Reject	10%
Cleveland	Reject	Reject	Phoenix	Fail to reject	Fail to reject
Dallas	Reject	10%	Pittsburgh	10%	10%
DC	Fail to reject	Fail to reject	Portland	Fail to reject	Fail to reject
Detroit	Fail to reject	Fail to reject	Saint Louis	Fail to reject	Fail to reject
Houston	Fail to reject	Fail to reject	Salt Lake City	Fail to reject	Fail to reject
Las Vegas	Fail to reject	Fail to reject	San Diego	Fail to reject	Fail to reject
Los Angeles	Reject	Fail to reject	San Francisco	Fail to reject	Fail to reject
Louisville	10%	10%	Seattle	Fail to reject	Fail to reject
Memphis	Fail to reject	Fail to reject	Tampa Bay	Reject	Reject

Note(s): “Reject” means rejection of the null hypothesis of a unit root at the 5% level, while “10%” means the null can be rejected at the 10% but not 5% level

Source(s): Author’s own work

Table 5. ADF and PP tests on rents

City	ADF test result	PP test result	City	ADF test result	PP test result
Atlanta	Reject	Reject	Miami	Reject	Reject
Baltimore	10%	10%	Minneapolis	Fail to reject	10%
Boston	Reject	Fail to reject	Nashville	Reject	Reject
Charleston	Reject	10%	New Orleans	Reject	Reject
Chicago	Reject	Reject	New York	Reject	Reject
Cincinnati	Reject	10%	Philadelphia	Reject	Reject
Cleveland	Reject	10%	Phoenix	Reject	10%
Dallas	Reject	Reject	Pittsburgh	Reject	10%
DC	Reject	Reject	Portland	Reject	Reject
Detroit	Fail to reject	Fail to reject	Saint Louis	Reject	Reject
Houston	Reject	Reject	Salt Lake City	Reject	Reject
Las Vegas	Reject	Reject	San Diego	Reject	Reject
Los Angeles	Fail to reject	Fail to reject	San Francisco	Fail to reject	Fail to reject
Louisville	Fail to reject	Fail to reject	Seattle	Reject	Reject
Memphis	Reject	Reject	Tampa Bay	Fail to reject	Fail to reject

Note(s): “Reject” means rejection of the null hypothesis of a unit root at the 5% level, while “10%” means the null can be rejected at the 10% but not 5% level

Source(s): Author’s own work

However, results in Table 5 show decidedly different results for rents. With the ADF test, we fail to reject at the 15% level in only six cases and fail to reject at the 10% level in only one other. With the PP test, we also fail to reject at the 5% level in just six cases and fail to reject at the 10% level in seven others. Thus, by both tests, the rental rates in most cities appear stationary, while prices do not. This rules out testing for cointegration. We will thus test for unit roots in the house price/rent ratio, as have previous authors. As previous studies have used panel unit root tests, we will do so as well. Note that for brevity of presentation, we have placed figures of the price-rent ratio for each of the 30 cities in the supplementary.

We begin with the Levin–Lin–Chu test. As this test requires a balanced panel, we include only the 13 cities for which data on prices and rents for all years from 1890 to 2006 were available. Given that Supplementary Figures 1–30 indicate an upward trend for all ratios, we include trends in all ratios for the test. As shown in Table 6, we cannot reject the null hypothesis that all series have unit roots in favor of the alternative that all series are stationary at any standard level. As the alternative hypothesis may be overly restrictive, and there may be cross-sectional dependence between the cities which can hamper inference, we apply the CIPS test to the data. We also include trends in this test. For comparison purposes, we first include just the 13 cities which we included in the Levin–Lin–Chu test. As displayed, we can reject the null that all ratios have unit roots against the less restrictive alternative that at least one ratio is stationary at less than the 1% level. As the CIPS does not require a balanced panel, we next conduct the test on all 30 cities. We obtain, as displayed at the bottom of Table 6, the same result as before, rejecting the null that all ratios have unit roots at less than the 1% level. Thus it appears there is a long-run relationship between prices and rents for at least some cities.

Although panel unit root tests have more power than their univariate counterparts, the results of Table 6 do not indicate which or how many cities have a stable long-run price-rent relationship. We thus apply standard univariate unit root tests to the price-rent ratio in Table 7. We include trends (and constants) in all test equations. With the ADF test, we are only able to reject the null of a unit root at the 5% level in three cases and at the 10% level in one other case. With the PP test, we can reject at the 5% level in six cases. Overall, these results cast doubt on the existence of a long-run relationship between prices and rents. However, ADF and PP tests can have low power, especially when trends are included in the test equations. The Elliott, Roth and Stock DF-GLS test addresses this issue by detrending the series’ first and then applying the unit root test, (potentially) increasing the power of the

Table 6. Panel unit root test results for the price-rent ratios

Levin–Lin–Chu test (13 cities)	Test statistic	<i>p</i> -value
	3.63	0.9999
CIPS test (13 cities)	–3.84735	<0.01
CIPS test (30 cities)	–3.562	<0.01

Note(s): The Levin–Lin–Chu test requires a balanced panel, so we restrict the sample in the test to the 13 cities where rent and house data are available for the full 1890–2006 sample (Atlanta, Boston, Chicago, Cincinnati, Dallas, Detroit, Los Angeles, Louisville, Minneapolis, Nashville, New York, San Francisco and Saint Louis). For comparison, we also perform the CIPS panel unit root test with the same 13 cities, and then perform the CIPS test with all 30 cities

Source(s): Author’s own work

Table 7. ADF and PP tests on the price-rent ratio

City	ADF test result	PP test result	City	ADF test result	PP test result
Atlanta	10%	Reject	Miami	Fail to reject	Fail to reject
Baltimore	Fail to reject	Fail to reject	Minneapolis	Fail to reject	Fail to reject
Boston	Fail to reject	Fail to reject	Nashville	Fail to reject	Reject
Charleston	Fail to reject	Fail to reject	New Orleans	Fail to reject	Reject
Chicago	Fail to reject	Fail to reject	New York	Fail to reject	Fail to reject
Cincinnati	Fail to reject	Fail to reject	Philadelphia	Fail to reject	Fail to reject
Cleveland	Fail to reject	Fail to reject	Phoenix	Fail to reject	Fail to reject
Dallas	Reject	Reject	Pittsburgh	10%	10%
DC	Fail to reject	Fail to reject	Portland	Fail to reject	Fail to reject
Detroit	Fail to reject	Reject	Saint Louis	Fail to reject	Fail to reject
Houston	Reject	Reject	Salt Lake City	Fail to reject	Fail to reject
Las Vegas	Fail to reject	Fail to reject	San Diego	Fail to reject	Fail to reject
Los Angeles	Fail to reject	Fail to reject	San Francisco	Fail to reject	Fail to reject
Louisville	Fail to reject	Fail to reject	Seattle	Fail to reject	Fail to reject
Memphis	Fail to reject	10%	Tampa Bay	Reject	Reject

Note(s): “Reject” means rejection of the null hypothesis of a unit root at the 5% level, while “10%” means the null can be rejected at the 10% but not 5% level

Source(s): Author’s own work

Table 8. DF-GLS test results on the price-rent ratio

City	Test result	City	Test result
Atlanta	Fail to reject	Miami	Fail to reject
Baltimore	Fail to reject	Minneapolis	Fail to reject
Boston	Fail to reject	Nashville	Fail to reject
Charleston	Fail to reject	New Orleans	10%
Chicago	Fail to reject	New York	Fail to reject
Cincinnati	Fail to reject	Philadelphia	Fail to reject
Cleveland	Fail to reject	Phoenix	Fail to reject
Dallas	Reject	Pittsburgh	Fail to reject
DC	Fail to reject	Portland	Fail to reject
Detroit	Fail to reject	Saint Louis	Fail to reject
Houston	Fail to reject	Salt Lake City	Fail to reject
Las Vegas	Fail to reject	San Diego	Fail to reject
Los Angeles	Fail to reject	San Francisco	Fail to reject
Louisville	Fail to reject	Seattle	Fail to reject
Memphis	Fail to reject	Tampa Bay	Reject

Note(s): “Reject” means rejection of the null hypothesis of a unit root at the 5% level, while “10%” means the null can be rejected at the 10% but not 5% level

Source(s): Author’s own work

test. We thus apply this test to the ratios. As displayed in [Table 8](#), we still fail to reject the null of a unit root in all but two cases at the 5% level and in one other at the 10% level.

All of these unit root tests impose parameter constancy, or the lack of a structural break, and of course unit root tests have low power in the presence of breaks. And given the changes in US housing markets over the past century, failure to allow for breaks is an omission. We thus run the Zivot–Andrews test on the ratio, with results in [Table 9](#). Although we have allowed for breaks

Table 9. Zivot–Andrews test results on the price-rent ratio

City	Test result	Break	City	Test result	Break
Atlanta	Reject	1918	Miami	Fail to reject	1991
Baltimore	Reject	1973	Minneapolis	Fail to reject	1989
Boston	Fail to reject	1981	Nashville	10%	1989
Charleston	Fail to reject	1989	New Orleans	Reject	1916
Chicago	Fail to reject	1986	New York	Fail to reject	1983
Cincinnati	Reject	1935	Philadelphia	10%	1935
Cleveland	Reject	1934	Phoenix	Fail to reject	1992
Dallas	10%	1930	Pittsburgh	Reject	1936
DC	Fail to reject	1917	Portland	Fail to reject	1975
Detroit	Reject	1933	Saint Louis	Fail to reject	1988
Houston	Reject	1918	Salt Lake City	Fail to reject	1943
Las Vegas	Fail to reject	1996	San Diego	Fail to reject	1991
Los Angeles	Fail to reject	1984	San Francisco	Fail to reject	1947
Louisville	Reject	1936	Seattle	Fail to reject	1992
Memphis	Fail to reject	1928	Tampa Bay	Fail to reject	1947

Note(s): The “Break” model was used, with one break allowed over the sample. “Reject” means a rejection of the null hypothesis of a unit root at the 5% level, while “10%” means the null can be rejected at the 10%, but not 5% level

Source(s): Author’s own work

under the alternative of stationarity in this case, we still cannot reject the null hypothesis of a unit root in most cases. Indeed, in 18 of 30 cases, we cannot reject a unit root at any standard level. In three cases we can reject at the 10% level, and in only nine cases do we reject at the 5% level. As discussed, the failure to allow for a break under the null distorts the timing of the break and thus can cause inference problems in testing for a unit root using the Zivot–Andrews

Table 10. Lee–Strazicich test results on the price-rent ratio

City	Test result	Break	City	Test result	Break
Atlanta	Reject	1945	Miami	Fail to reject	1957
Baltimore	Reject	1954	Minneapolis	Fail to reject	1981
Boston	Reject	1967	Nashville	10%	1932
Charleston	10%	1965	New Orleans	Reject	1945
Chicago	Reject	1993	New York	Reject	1979
Cincinnati	Reject	1939	Philadelphia	Reject	1926
Cleveland	Reject	1955	Phoenix	10%	1955
Dallas	Reject	1928	Pittsburgh	Reject	1953
DC	Reject	1931	Portland	10%	1966
Detroit	Reject	1933	Saint Louis	Reject	1964
Houston	10%	1910	Salt Lake City	10%	1917
Las Vegas	Fail to reject	1937	San Diego	10%	1971
Los Angeles	Reject	1929	San Francisco	Reject	1944
Louisville	10%	1956	Seattle	Fail to reject	1959
Memphis	Reject	1932	Tampa Bay	Reject	1955

Note(s): The “Break” model was used, with One break allowed over the sample. “Reject” means a rejection of the null hypothesis of a unit root at the 5% level, while “10%” means the null can be rejected at the 10%, but not 5% level

Source(s): Author’s own work

method. We therefore apply the Lee–Strazicich test, which allows for breaks under both the null and alternative, to the 30 ratios, with results displayed in Table 10.

We first note that the timing of breaks is highly varied, with no real pattern. However, the key striking take from these results is that in most cases, unlike with the ADF, PP, ERS and Zivot–Andrews tests, there does appear to be a long-term relationship between rents and prices. In 18 of 30 cases, we can reject the null of a unit root at the 5% level, and in eight other cases at the 10% level. Only for Las Vegas, Miami, Minneapolis and Seattle do we fail to reject the null hypothesis of a unit root at any standard level. There does not appear to be a discernible pattern. While Las Vegas and Miami were clearly among the hardest-hit markets over the 2007–2009 Global Financial Crisis, Seattle and Minneapolis experienced price declines that were not as wrenching as those in other cities. Overall, it does appear that prices and rents, at least in most of these 30 cities, share a long-run relationship.

User costs, interest rates, mortgage credit and breaks

Recall that while some models have been developed which imply house prices and rents should have a stable, stationary bivariate long-run relationship, there are other models (Himmelberg *et al.*, 2005) which specify the price-rent ratio as the inverse of the user cost of housing, which includes the long-term real interest rate, risk premium, taxes and the expected capital gain from owning the house. Amaral *et al.* (2024) found an inverse relationship between interest rates and the difference in the geographic variation in house prices versus rents in the USA over 1980–2018, and this difference has been growing over time. The fact that this difference in house price dispersion versus rent dispersion has been growing may seem, at first glance, contrary to our finding that there appears, for most cities, to be a stationary long-run relationship between prices and rents, a point we will address below.

We would like to include user cost in our analysis. However, it is not feasible to obtain expected capital gains across all of these cities, and there are other assumptions regarding risk premia and tax considerations that are also hard to get reliable measures for, so we will not perform a “structural” analysis. We will, however, examine the rough interaction of prices, rents interest and credit availability and how these relationships may have changed over time.

First, as we are examining housing over 1890–2006, we gather data from the Macrohistory Database, (www.macrohistory.net/database/), which was compiled by Jorda, Knoll, Kuvshinov, Schularick and Taylor. This site has data for numerous countries, including the USA, on long-term interest rates, inflation and credit. Given the relationship found by Himmelberg *et al.* (2005), Hill and Syed (2016) and Amaral *et al.* (2024) between long-term interest rates and the price-rent relationship, we obtain nominal long-term rates from this site for the USA over 1890–2006 to match with our house price and rent data. We also obtain data on the CPI for the USA over these years. Given the emphasis by Himmelberg *et al.* (2005) on using real, rather than nominal interest rates in analyzing user cost equilibrium, we calculate annual inflation with the CPI data, and deduct it from nominal long-term interest rates to obtain real long-term rates.

The financing of housing purchases has, of course, changed over the decades, and the availability of credit can have an impact on how house prices, rents and interest rates interact. We thus obtain nominal mortgage credit, as well as nominal GDP from the Macrohistory database, and divide the former by the latter to obtain mortgage credit relative to output. Graphs of real interest rates and mortgage debt to GDP are displayed in Figures 6 and 7.

First, examining breaks in Table 10, we find most breaks (21 of 30) in the price-rent ratio occur from the 1930s to the 1960s. Does this correspond to large changes in interest rates? In



Figure 6. Real long-term interest rates
Source: Author's own work

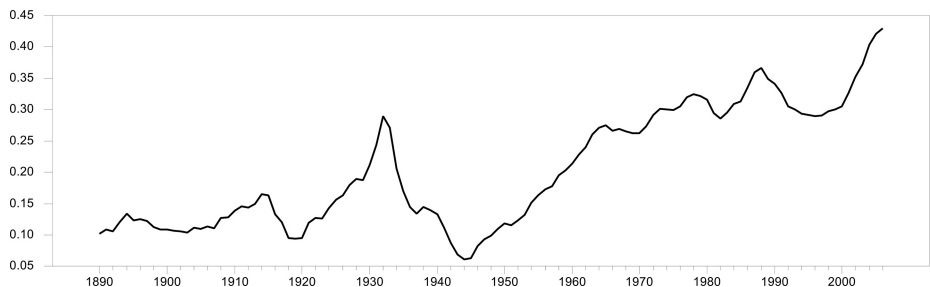


Figure 7. Mortgage debt/GDP
Source: Author's own work

Figure 6, there was a major shift in real interest rates from a peak in 1932, followed by a prolonged drop until the late 1940s, and a subsequent recovery until the early 1970s. It is true that there was an even sharper decrease, increase and decline over 1915–1926, but this involved First World War, and the change from the 1930s to the 1970s was more prolonged. Thus there does appear to be some relationship between large changes in interest rates and structural breaks in the price-rent relationship.

As displayed in Table 9, we also conducted the Zivot–Andrews test. While we have noted that the method of detecting breaks with the Zivot–Andrews method can lead to erroneous break dates, resulting in mistaken unit root test results (Lee and Strazicich, 2013), we did obtain estimated break dates. Table 9 shows these dates. While there were some (10 of 30) breaks over the 1930s–1960s, when most breaks were found for the Lee–Strazicich technique, we found numerous breaks in the 1980s (8 of 30 cases) with the Zivot–Andrews, while we found only one city with a break in the 1980s with the Lee–Strazicich method. While we believe the Lee–Strazicich method is more reliable, we do note that the 1980s were a time in which interest rates fell from a high local peak and began a secular decline (Figure 6). In addition, as discussed below, the 1980s were a time of major regulatory changes in housing finance. In particular, the end of Regulation Q in 1980 meant more funds would be made available for mortgages, and the growth of securitization also led to the growth of housing credit availability. And we also develop an aggregate measure of the price-rent ratio, which, as will be discussed, exhibits a break at 1980.

We note that, although the Zivot–Andrews and Lee–Strazicich tests are used both to determine whether a series has a unit root and examine whether there is structural change, there are other methods of testing for breaks. The Bai–Perron test is applied to time series to estimate when endogenous breaks may have occurred. There is an implicit assumption of stationarity when applying the test. We also apply the test to the 30-city price-rent ratios, and display the results in [Table 11](#). As shown, there are significant breaks in 26 of 30 cities. While there were some breaks (8 of 30) from the 1930s to 1960s, the same period when the Lee–Strazicich method detected the most breaks, there were more breaks estimated in the 1980s with the Bai–Perron method (7 of 26 versus 1 of 30). Given that, even with the Lee–Strazicich test, there are still some cities with nonstationary price-rent ratios, we believe the Lee–Strazicich method is the more appropriate test for determining breaks in for this data. But again, there are some breaks, with the Bai–Perron method, in the 1980s.

Of course, each of the 30 cities will reflect some idiosyncratic factors, so we create an aggregate price-rent ratio by summing the ratio for each of the 13 cities (Atlanta, Boston, Chicago, Cincinnati, Dallas, Detroit, Los Angeles, Louisville, Minneapolis, Nashville, New York, San Francisco and Saint Louis) for which data was available for the full 1890–2006 sample. This aggregate measure of the price-rent ratio is displayed in [Figure 8](#). As displayed, this ratio did fall and rise from the 1930s to the end of the 1960s (although the clearest, sharpest increase was from the end of the 1960s to the end of the sample, when interest rates both rose sharply and also fell) when interest rates fell and also rose, and there were a large number of structural breaks for individual cities.

To further examine the relationship between interest rates and the price-rent ratio, we acknowledge that as we cannot obtain data on expected capital gains, we cannot perform a formal structural analysis. However, just as [Davis and Heathcote \(2007\)](#) performed simple regressions of house prices and land on interest rates, we too will run some simple models of the ratio on long-term real rates, as well as on mortgage credit.

In [Table 12](#), we see that there is a clear *positive* relationship between the interest rate and the price-rent ratio over the entire 1890–1906 sample. While this result appears to contradict those of [Amaral et al. \(2024\)](#), note that there is no control for other factors, such as credit. Greater

Table 11. Bai–Perron test results on the price-rent ratio

City	Break	City	Break
Atlanta	1948	Miami	1933
Baltimore	1974	Minneapolis	1990
Boston	1984	Nashville	1940
Charleston	1988	New Orleans	1915
Chicago	1990	New York	1989
Cincinnati	1935	Philadelphia	1990
Cleveland	1934	Phoenix	1993
Dallas	1930	Pittsburgh	1982
DC	NA	Portland	1983
Detroit	1931	Saint Louis	1980
Houston	1918	Salt Lake City	1921
Las Vegas	1999	San Diego	1992
Los Angeles	NA	San Francisco	NA
Louisville	NA	Seattle	1993
Memphis	1981	Tampa Bay	1949

Note(s): NA refers to the lack of a significant structural break

Source(s): Author's own work

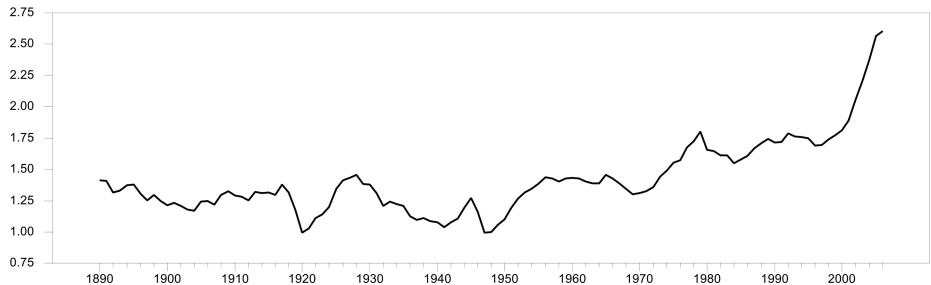


Figure 8. Aggregate price/rent ratio

Note(s): This is the house price-rent ratio for the 13 US cities (Atlanta, Boston, Chicago, Cincinnati, Dallas, Detroit, Los Angeles, Louisville, Minneapolis, Nashville, New York, San Francisco and Saint Louis) for which data for the whole 1890–2006 time span is available. Note that there is a break by the Lee–Strazicich test in 1980. This coincides with the end of Regulation Q, which had constrained credit for mortgages. It also marked the beginning of securitization (Pozdena, 1990). Both of these changes made credit for housing more available, possibly contributing to the rise in the house price/rent ratio

Source: Author’s own work

Table 12. Regression results

Full sample	Interest rate	Mortgage credit
Coefficient	0.011944	
T-stat	1.99	
Coefficient		2.588
T-stat		16.08
Coefficient	−0.005052	2.66
T-stat	−1.43	15.83
1980–2006 sample		
Coefficient	−0.1004	
T-stat	−4.78	
Coefficient		5.75
T-stat		7.35

Note(s): These are the results of the regression of the aggregate price-rent ratio on the real long-term interest rate (first rows), then a regression of the ratio on mortgage credit as a fraction of GDP (second set of rows) and a regression ratio on both the interest rate and mortgage credit (third set of rows), all of the above over the entire 1890–2006 sample. In the last two sets of columns are results of Two univariate regressions of the ratio on the interest rate and mortgage credit

Source(s): Author’s own work

credit availability may change the association of interest rates and the ratio, and there thus may have been changes later in the sample, especially since 1980. The post-1980 years are the sample used by Amaral *et al.* (2024) and were also a period in which more credit was available, not just in terms of the amount of mortgage credit extended by traditional lenders but also from other investors through the process of securitization (Pozdena, 1990; Miles, 2019).

As credit could have an impact on the ratio, we first examine the evolution of mortgage credit to GDP over 1890–2006. As displayed in Figure 7, mortgage credit was very low early in the sample but then started rising around 1907 until First World War and then fell and then rose

sharply from 1920 to the Great Depression, fell again, and has been exhibiting an overall strong upward trend since the end of Second World War. A simple regression of the price-rent ratio on mortgage credit, displayed in Table 12, shows a positive relationship between the two variables.

We then regress the aggregate price-rent ratio on both the long-term real rate and mortgage credit as a portion of GDP. As shown in Table 12, mortgage credit is still positive and significant, while interest rates have a negative, though, insignificant impact on the ratio. Again, the relationship between interest rates and the ratio may have changed over time.

To see if there may have been a break in the dynamics of the ratio and its relationship with interest rates and credit, we first test for the stationarity of the aggregate 13-city ratio. As shown in Table 13, we are unable to reject the null of a unit root with the ADF test. We then allow for a break with the Zivot–Andrews method, but as displayed, we cannot reject the null of a unit root at the five or even 10% level. However, when we use the more appropriate Lee–Strazicich test, we can reject the null of a unit root at the 5% level. This result is similar to that for the majority of individual city ratios. Moreover, there is a break at 1980, which coincides with the start of Amaral *et al.*’s analysis. 1980 also corresponds to a large change in interest rates. Long-term real borrowing costs went from a trough in 1974 to a peak in 1984 and began a decline lasting to the end of the sample.

In addition, 1980 marked a change in financial regulation which made financing of housing purchases more available. Regulation Q was abolished that year. This regulation prevented depository banks in the USA – then the main source of mortgages – from offering interest on checking accounts, which is a key source of funding for housing loans. This meant that, if the economy was growing and pushing up housing demand, and interest rates started to rise, depositors would withdraw funds from banks, depriving the lenders of funds for mortgages, and putting a brake on housing demand. Once Regulation Q was abolished, rising interest rates would no longer deprive banks of funds for mortgages, and lending could proceed. In addition, the early 1980s was also a time when securitization developed and grew, which meant that there were new funds available for home loans from investors, both in the USA and abroad (see Pozdena, 1990 and Miles, 2019) for a discussion of these changes). These changes could, of course, change the relationships among the ratio, interest rates and credit.

We thus seek to see how the relationships among interest rates, credit and the ratio have changed since the breakpoint at 1980. As shown in Table 12, we regress the aggregate ratio on the interest rate and then in a separate simple regression on mortgage credit, restricting the

Table 13. Unit root test results for the aggregate price-rent ratio

Test method	Result
ADF	Fail to reject
DF-GLS	Fail to reject
Zivot–Andrews	Fail to reject
Break date	1936
Lee–Strazicich	Reject
Break date	1980
Bai–Perron break date	1976

Note(s): As displayed, for no test could the null hypothesis of a unit root be rejected at the 5% (or 10%) level, except in the case of the Lee–Strazicich test, where the null was rejected at the 5% level. There are break dates for the Zivot–Andrews and Lee–Strazicich tests. Although it was not a test for a unit root, we also estimated an autoregressive model and tested for a break with the Bai–Perron technique

Source(s): Author’s own work

sample to 1980–2006. The interest rate now has an unambiguously negative association with the price-rent ratio. Thus, with the advent of greater credit, there is a clear negative association between interest rates and the ratio, as in [Himmelberg et al. \(2005\)](#) and [Amaral et al. \(2024\)](#). In addition, as [Table 12](#) displays, the association between credit and the ratio is also greater after 1980 than over the whole sample.

As [Amaral et al. \(2024\)](#) have found a decrease in the interest rate associated with an increase in the dispersion of prices compared to that of rents (and presumably an increase in the house price-rent ratio), while we have found a trend-stationary relationship with a structural break for the ratio, our results and those of [Amaral et al.](#) would seem at first glance contradictory. However, there is no contradiction in the two sets of results.

It is true that, since 1980, interest rates have been negatively associated with the price-rent ratio. It is also the case that over this period, the ratio has increased in most cities (see [Figure 8](#) on the aggregate ratio). These developments are consistent with the results of [Amaral et al. \(2024\)](#). However, it is also true that despite appearing nonstationary by the ADF and Zivot–Andrews tests, there does appear to be a trend-stationary relationship between home prices and rents. Again, this relationship includes a trend, as well as a break, so the price-rent ratio is not constant and has been rising on average through time. It is also calculated over a 116-year sample, rather than a sample of 30–40 years. As noted, [Ambrose et al. \(2013\)](#) stated that a problem with previous studies on the topic is a lack of long-run data. [Ambrose et al.](#) did find a long-run relationship between prices and rents for Amsterdam but used a 355-year sample to do so.

Policy recommendations

We again note that the break for the aggregate home price-rent ratio occurred in 1980, which coincides with, among other changes, the abolition of Regulation Q and the growth in securitization ([Pozdena, 1990](#)). As the ratio, though stationary, appeared to grow from that point, it seems that greater credit has more of an impact on home prices than apartment rents. This may be owing to the greater land intensity of houses compared to apartments ([Yao, 2021](#)).

For commercial applications, note there are some REITs which specialize only in apartments. Although house prices and rents exhibit a stationary relationship, the fact that home values have become more geographically varied compared to rents in this sample (although this difference in variability is less than that found in studies using more limited data sets, such as [Tang et al. \(2020\)](#) and [Yao \(2021\)](#)), it remains the case that REITs which focus exclusively or even mainly on apartments could obtain portfolio diversification benefits by adding some houses to their asset holdings, as there is a greater variability across the country in home prices than rents.

Another issue that growing house-price/rent ratios may create is a reduction in labor mobility. Workers who desire to be homeowners may be prevented from moving from one region to another, as they must pay substantially more to buy a home compared to rental costs. This could both lower wage gains and employment opportunities for workers as well as lower total GDP for the economy ([Hsieh and Moretti, 2019](#)).

In terms of policy, we have found that lower interest rates and greater credit seem to be associated with a higher price-rent ratio. Thus policies designed to bring rents and prices into greater parity should tend to be local, as credit and interest policies are set nationally. One obvious policy is zoning restrictions. [Glaeser and Gyourko \(2003\)](#) is one of numerous studies documenting how home prices rise in response to building regulations. Moreover, while we, with our small (30 city) sample, have not found a significant correlation between housing supply elasticity and the ratio, it is likely that zoning laws affect house prices more than apartment rents. This is owing to the greater land

intensiveness of apartments. Restricting land use will thus disproportionately affect house prices compared to apartment rents. Thus, allowing for greater home-building can lower the ratio of prices and rents.

Data and code availability statement

As noted, the data for house prices and rents in this paper was obtained from the Philadelphia Federal Reserve Bank at: www.philadelphiafed.org/surveys-and-data/regional-economic-Manalysis/historical-housing-prices that was developed as part of a project to develop more accurate data on home prices, and especially rents, than had previously been available. Data for long-term interest rates and the CPI (which were used to calculate long-term real interest rates, as shown in Figure 6) and mortgage credit to GDP were obtained from the Jorda–Schularick–Taylor Macrohistory database at www.macrohistory.net/database/. All tests in this paper were performed with the EViews statistical software package.

Conclusion

Several previous studies failed to find a stationary relationship between home values and rents. We have done so with a new, much longer data set and with a test that allows for structural breaks. There are some caveats. First, the existence of a time trend in the tests, given the obviously trending nature of the ratios, means that the price-rent relationship is trend stationary but not constant. Over the more than 100 years of the sample, prices have risen more than rents. Although the difference in geographic dispersion between prices and rents is not as great as that found by Yao (2021), the fact that apartments take up less space than houses may have led to house prices rising faster, on average, than rents over the decades. Note that the increase in the price-rent ratio for housing is not dissimilar to the increase in the price/earnings ratio for equities.

In addition, the fact that we failed to reject the null of a unit root with standard tests for the ratio in most cities indicates that although the ratio is stationary, prices can deviate from this trend-stationary relationship with rents for prolonged periods. However, it is clear that home prices generally do not become unmoored over the (very) long run from their fundamental determinant, rents.

We do note that while this data is based on hedonic indices created from 2.7 million newspaper listings, making it much more reliable than price and rent data heretofore available (quality-adjusted rent data was not available for the USA before 2006), it does end in 2006. It would be interesting to consider the implications of the analysis for the nearly two decades since.

We first see, in Figure 8, that the average price-rent ratio has been rising over time, especially since the 1980 break. However, it is also the case that when allowing for a trend and a break, we find that the ratio, despite rising over time due to the trend and break, is still trend-stationary over our long 116-year sample. While further data from the Agorastos *et al.* (2024) project is not available post-2006, there are indications that prices are continuing to outpace rents in the USA. We have noted that Amaral *et al.* (2024), using more standard data than Agorastos *et al.* (2024), find the price-rent ratio has continued to rise over their 1980–2018 sample. Again, this does not imply a nonstationary relationship for prices and rents in the USA. However, it does suggest, given the trend and break, that homeownership, relative to renting, has become harder to achieve in the USA and that this trend has been continuing since 2006.

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Supplementary material

The supplementary material for this article can be found online.

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